

Enhancement mechanism of ultrasonic vibration on interface bonding strength in laser joining of polymer to ceramic

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ABSTRACT

The current challenge in laser joining polymer and ceramic heterogeneous structures lies in their relatively low mechanical strength, primarily attributed to weak interface bonding. To address this issue, this paper proposes the application of ultrasonic vibration during the laser joining process of PET with both Si₃N₄ and Ti-coated Si₃N₄ sheets to enhance the interface bonding strength. Investigating the mechanisms underlying the enhancement of interface bonding strength due to ultrasonic vibration is crucial for improving the overall mechanical strength of the joints. It is found that the physical wetting thermodynamic conditions and chemical bond thermodynamic conditions for laser joining of PET to ceramics were the surface tension of ceramics higher than that of the molten PET and the variation in the Gibbs function less than -24.3 kcal/mol, respectively. Experimental results demonstrated that the bonding interface between Si₃N₄ sheet and PET exhibits only physical and mechanical bonding, with no evidence of new chemical bonds. However, XPS analysis detected new Ti—C bonds at the bonding interface of Ti-coated Si₃N₄ and PET. Furthermore, experimental results substantiated that ultrasonic vibration could improve the spreading wetting of PET melt on the Si₃N₄ surface, resulting in a 17 % increase in the bonded area of the weld seam. Additionally, ultrasonic vibration enhanced the molten PET filling into the blind hole structure on the Si₃N₄ surface, resulting in an approximately 73 % increase in the filling depth of the blind holes. Lastly, ultrasonic vibration facilitated the chemical reaction between the molten PET and Ti atoms, resulting in an approximately 57 % increase in the thickness of the interface containing Ti—C bonds. It is evident that ultrasonic vibration can bolster the interfacial bonding strength of PET/Si₃N₄ and PET/Ti-coated Si₃N₄ in laser joining. Moreover, in comparison to physical bonding, its more pronounced impact on mechanical and chemical bonding is noteworthy.

1. Introduction

In recent years, hybrid components formed from medical-grade polymers and bioceramic materials have been widely employed in the field of biomedicine, including applications such as bioreactors, lab-on-a-chip systems, and implantable medical devices [1–4]. Utilizing both bioceramic materials and medical polymers in medical products allows for the realization of the high strength and excellent wear resistance of ceramics, while also taking advantage of the excellent biocompatibility and elasticity of medical-grade polymers [1,3,5]. However, a major obstacle remains: the low mechanical strength of the joints connecting these dissimilar materials, which necessitates the development of reliable and robust joining techniques [1–3].

Traditional joining methods such as adhesive bonding and mechanical fastening have limitations like poor fatigue life and unsatisfactory sealing capabilities [6,7]. While newer techniques like co-cured layer stacking [8], friction stir welding [9,10], and laser joining [4,11–13] have been investigated, these approaches have their own set of shortcomings. For instance, co-cured layer stacking suffers from the risk of delamination [8], and friction stir welding is unsuitable for complex joint geometries and confined spaces [14,15].

Laser joining has shown promise due to its numerous advantages such as minimal heat-affected zones, low stress concentration, and high automation [16]. However, research into laser joining of polymer to ceramic hybrid joints remains in its nascent stage, with challenges including low tensile and shear strength of the resultant joints

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[4,11–13]. For instance, Y. Kawahito achieved a joint between polyethylene terephthalate (PET) and Si_3N_4 ceramic using a specific wavelength diode laser; however, the resulting tensile strength was an underwhelming 4.7 MPa—attributable to weak interfacial bonding that led to peeling failures [11]. In a similar vein, K.F. Tamrin's work with CO_2 laser-based heat conduction welding yielded a PETG-Macor glass joint, but disappointingly, even under the best circumstances, the shear strength barely crossed the 6.3 MPa mark [4,12]. Liu and colleagues ventured into laser transmission welding with semiconductor nanosecond lasers to join Al_2O_3 ceramic and PMMA. Though some improvement was observed through mechanical riveting enabled by molten PMMA flowing into micro-blind holes on the ceramic surface, the maximum joint strength stalled at around 25 MPa [13].

To address the current challenges in laser joining of polymer-ceramic hybrid structures, this paper introduces an innovative approach: the incorporation of ultrasonic vibrations during the laser joining process. This method has also been applied to study laser joining of polymer to metal hybrid structures, demonstrating the positive impact of ultrasonic vibrations on tensile shear strength by improving interface bonding strength [17–19]. The primary goal of this study is to gain a deeper understanding of how ultrasonic vibrations enhance physical bonding, mechanical anchoring, and chemical bonding at the polymer-ceramic interface, thus significantly improving joint strength.

2. Experimental method

2.1. The ultrasonic-aided laser joining

Fig. 1 presents a schematic of the experimental apparatus used for the ultrasonic-aided laser joining of ceramic to polymer (U-LACP) system. The setup features three main components: a Nd:YAG pulsed laser (HAN'S, PB300CE) serving as the heating source, an ultrasonic vibration system, and a clamping apparatus to hold the workpieces securely. The ultrasonic vibration device operates in the longitudinal mode and is capable of vibrating at a frequency of 28 kHz. This functionality is made possible by a specialized transducer tool in the ultrasonic system, where the workpiece is situated. For the U-LACP joining process, the PET sheet is laid over the ceramic sheet. The laser beam then passes sequentially through a layer of quartz glass and the PET sheet to reach and heat the ceramic surface. This heating process subsequently leads to the melting of the PET sheet through thermal conduction. Concurrently, the ultrasonic vibration system is engaged, operating at a duty cycle of 20 % as illustrated in Fig. 2. The specific joining parameters utilized in the experiment are detailed in Table 1.

2.2. Materials to be joined

The materials to be joined were biocompatible Si_3N_4 ceramic and polyethylene terephthalate (PET) sheets with dimensions of $60 \times 20 \times 2$ mm and $60 \times 20 \times 1$ mm, respectively. They were purchased from

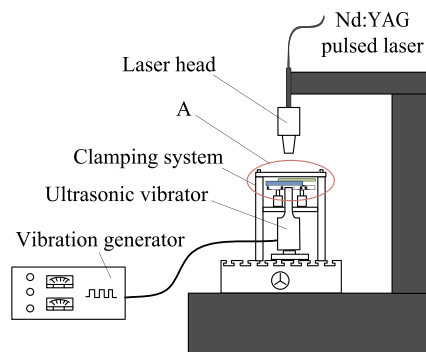


Fig. 1. The ultrasonic-aid laser joining of ceramic to plastic (U-LACP) system.

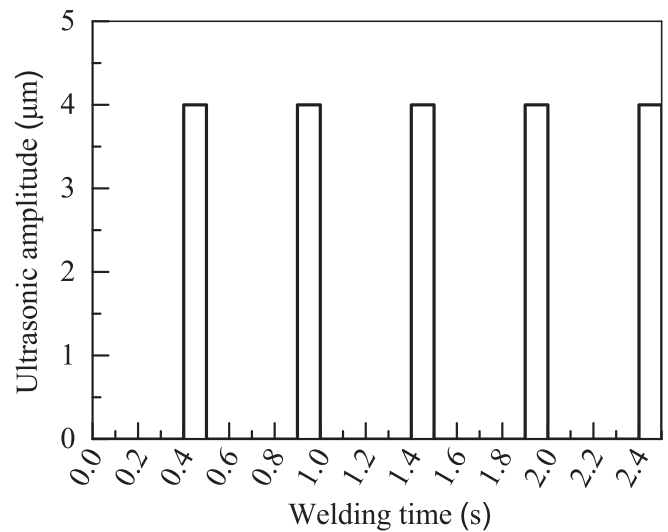


Fig. 2. The ultrasonic pulse operation mode (duty cycle of 20 %).

Table 1

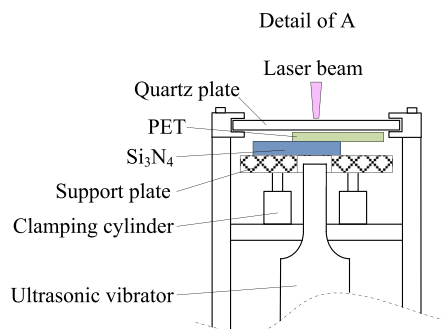
Joining parameters.

Laser average power	40 W
Laser pulse frequency	20 Hz
Laser spot diameter	φ4 mm
Laser focal length	25 mm
Joining speed	4 mm/s
Pulse duration	8 ms
Ultrasonic vibration amplitude	4 μm
Ultrasonic duty cycle	20 %
Clamping force	0.6 MPa

Goodfellow Cambridge Ltd.

An ultraviolet pulsed laser (HAN'S, EP-50-THG-S) was employed to fabricate a regularly-arranged microstructure of blind holes on the surface of Si_3N_4 . The blind holes machining experiments were conducted at room temperature. During the process, the laser beam traced concentric circles, moving inward from the outer edge in a circular rotational cutting motion. This approach successfully produced a structured array of blind holes in both the X and Y directions, with center-to-center spacings of 0.5 mm along the X direction and 1 mm along the Y direction, diameters approximately 100 μm, and depths around 1300 μm, as demonstrated in the following Section 3.3.

A thin titanium (Ti) film (about 2 μm thickness) prepared by magnetron sputtering coating technology was utilized on Si_3N_4 sheets to promote the joining with PET, where the Ti film play a role of coupling agent. The magnetron sputtering of Ti coating on Si_3N_4 sheets was fabricated by scia Magna 700 system (scia Systems) at a deposition rate



of 10 nm/min under argon condition.

The parent material of Si_3N_4 ceramic sheets was ground using diamond emery paper (#1500 and #2000) to ensure a uniform surface roughness. Concurrently, additional grades of diamond emery paper (#320, #1000, #1500, and #2000) were utilized to remove any sedimentary protrusions around the blind holes. The surfaces of all the samples, including the PET sheets, the Si_3N_4 sheets both with and without blind holes, Si_3N_4 sheets with a Ti coating, and Si_3N_4 sheets that underwent emery paper finishing, were ultrasonically cleaned in an alcohol bath for 10 min. Subsequently, they were dried at 50 °C for 12 h in preparation for laser joining.

2.3. Analysis of the joint

An optical microscope was utilized to record the appearance of bonding area. Concurrently, Scanning Electron Microscopy (SEM, ZEISS GeminiSEM300) was employed to investigate the morphological features of the joint interface. For the SEM analysis, longitudinal sections of the specimens were scrutinized. These specimen sections were prepared through mechanical grinding, followed by polishing with 1 μm diamond paste.

X-ray Photoelectron Spectroscopy (XPS, ULVACPHI 1800) was employed to investigate the chemical bonding characteristics at the joint interfaces. The bonded samples were mechanically separated at the interface through shear tests, and analytical evaluations were conducted on both the Ti-coated and uncoated Si_3N_4 portions of the joint. To unveil the nature of the chemical bonds at the interface, ion milling in incremental steps was utilized to thin down and eliminate any solidified PET adhered to both the Si_3N_4 and Ti-coated Si_3N_4 surfaces, until reaching the Si_3N_4 or Ti substrate, respectively. This procedure was validated by performing XPS analyses after the removal of each layer of PET, until either Si or Ti elements were detected. These layers were assigned depth levels denoted as $d = 0$ for the sample joints that were not subject to ultrasonic vibration, and $d_u = 0$ for those that were. At these specific depths (either $d = 0$ or $d_u = 0$), additional ion milling was performed to elucidate the relationship between depth and chemical bond intensity. Distances cited in this paper are measured from these baseline depths of $d = 0$ or $d_u = 0$. The detection positions and the procedural outline are illustrated in Fig. 3.

3. Results and discussion

3.1. Improvement of physical bonding at the PET/ Si_3N_4 hybrid joint interface

3.1.1. Thermodynamics conditions

Physical bonding is characterized as a specific form of cohesive interaction that is triggered by the attractive forces between atoms or molecules, notably intermolecular forces, which encompass van der Waals forces and hydrogen bonds. This mechanism is initiated when the separation distance between the surfaces of two targeted material substrates is confined to a diminutive range of 3–5 Å [20,21]. The formation of the interface between PET and Si_3N_4 is a complex physicochemical process, including melting, flow, wetting, diffusion and solidification of PET. The theoretical underpinning of the physical bonding between

Si_3N_4 and PET is grounded in surface energy theory, where the magnitude of the bonding force is contingent upon the wettability of the surfaces. Accordingly, scholars frequently utilize wettability as an evaluative metric to predict or assess the efficacy of the interfacial bonding [22–24]. Significantly, the surface of Si_3N_4 exhibits a proclivity for gas adsorption due to the influence of van der Waals forces, which include dipole forces, induced forces, and dispersion forces [24–26]. Under laser-activated bonding conditions, if the molten state of PET can sufficiently spread and wet the Si_3N_4 surface under the interactive forces of atoms, it concurrently displaces the adsorbed gases at the interface. This facilitates intimate surface-to-surface contact between Si_3N_4 and molten PET, thereby activating intermolecular forces—primarily van der Waals forces—to effectuate physical bonding.

Grounded in thermodynamic principles, the process under consideration involves the transformation of a “gas-solid” interface into both a “liquid-solid” and a “gas-liquid” interface upon the spreading and wetting of PET on a Si_3N_4 surface [23]. The actual surface of the wetted Si_3N_4 is denoted as A_S , and the surface area of the wetted PET melt is denoted as A_L . Then, the work W done by the system during the wetting process satisfies:

$$W = \gamma_S \cdot A_S - \gamma_L \cdot A_L - \gamma_{SL} \cdot A_S \quad (1)$$

In Eq. (1), γ_S represents the surface tension (surface energy) of Si_3N_4 , γ_L denotes the surface tension of PET and γ_{SL} signifies the interfacial tension between Si_3N_4 and the PET melt. Under isothermal and isobaric conditions, the change in the system's Gibbs free energy (ΔG) per unit area during the wetting process of Si_3N_4 by the PET melt is as follows:

$$-\Delta G = \frac{W}{A_S} = \frac{\gamma_S \cdot A_S - \gamma_L \cdot A_L - \gamma_{SL} \cdot A_S}{A_S} \quad (2)$$

The surface of Si_3N_4 is polished using #2000 diamond sandpaper, resulting in relatively low surface roughness, then $A_S \approx A_L$, thus:

$$\Delta G = \gamma_{SL} + \gamma_L - \gamma_S \quad (3)$$

In a liquid-solid contact system at equilibrium, the interrelationship among surface tension, interfacial tension, and the contact angle is formally articulated by Young's Equation [27],

$$\cos\theta = \frac{\gamma_S - \gamma_{SL}}{\gamma_L} \quad (4)$$

In Eq. (4), θ means the contact angle between PET and Si_3N_4 , which

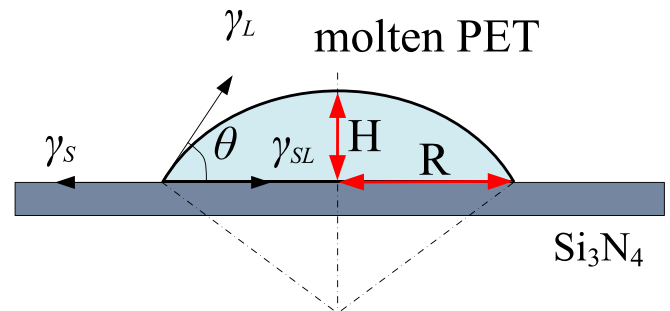


Fig. 4. Contact angle between the molten PET and Si_3N_4 .

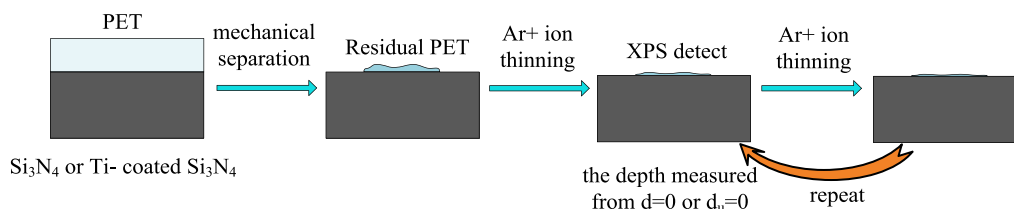


Fig. 3. The schematic diagram of XPS detection process.

is commonly used to quantitatively represent the wettability of a droplet on a solid substrate as shown in Fig. 4. When θ is less than 90° , the system is considered to be hydrophilic; otherwise, we call it hydrophobic. As illustrated geometrically in Fig. 4, the contact angle can be mathematically delineated as follows:

$$\theta = 2\arctan\frac{H}{R} \quad (5)$$

Substituting Young's Equation into Eq. (3) yields the following result:

$$\Delta G = \gamma_L(1 - \cos\theta) \quad (6)$$

When alteration of free energy of the whole system ΔG is negative, the PET melt can spontaneously spread and wet the surface of Si_3N_4 substrate. As indicated by Eq. (6), it is only under the conditions where $\theta = 0$ and $\cos\theta = 1$ that the critical thermodynamic conditions for spreading and wetting can be met. Through the measurement of different liquids with varying surface tensions γ_L and their corresponding contact angles θ on a consistent solid surface, Zisman [28] discovered that plotting $\cos\theta$ against γ_L yields an approximate linear relationship. By extrapolating this line to $\cos\theta = 1$, a characteristic parameter reflecting the surface properties of the solid, namely the critical surface tension γ_C can be determined. When surface tension of a particular liquid γ_L is less than or equal to this critical surface tension γ_C , the liquid will naturally spread and wet the solid surface. This phenomenon is corroborated by the equation governing thermodynamic equilibrium states [28],

$$\gamma_C = \gamma_S - \gamma_{SL} \quad (7)$$

It is found that the critical surface tension γ_C values derived using polar and non-polar liquids differ [29]. This discrepancy arises from variations in the interfacial tension γ_{SL} of the liquid. As γ_{SL} decreases, the γ_C value increases. From Eq. (7), it is evident that the maximum value of γ_C is equivalent to the surface tension of the solid γ_S . Thus, the fundamental condition for spontaneous spreading and wetting is: $\gamma_L \leq \gamma_C \leq \gamma_S$, implying that the liquid's surface tension γ_L must be less than the solid's surface tension γ_S . In essence, as long as the surface tension of Si_3N_4 γ_S exceeds that of the PET melt γ_L , the PET melt can spread and wet the Si_3N_4 , culminating in a physical bond.

The surface tension of PET melt is around 0.027 J/m^2 [30]. And, the surface tension of Si_3N_4 is around 2.583 J/m^2 (Biolin, Sigma 700/701), which is far greater than 0.027 J/m^2 . Thus theoretically, the contact angle θ between PET melt and Si_3N_4 is less than 90° , and PET melt is able to spread and wet the Si_3N_4 and thus they form physical bonding.

3.1.2. Improvement of wetting performance

The cross-sectional morphology of the interface between the PET and Si_3N_4 was observed and analyzed by the SEM as shown in Fig. 5. It is revealed that the PET melt successfully wets the Si_3N_4 surface,

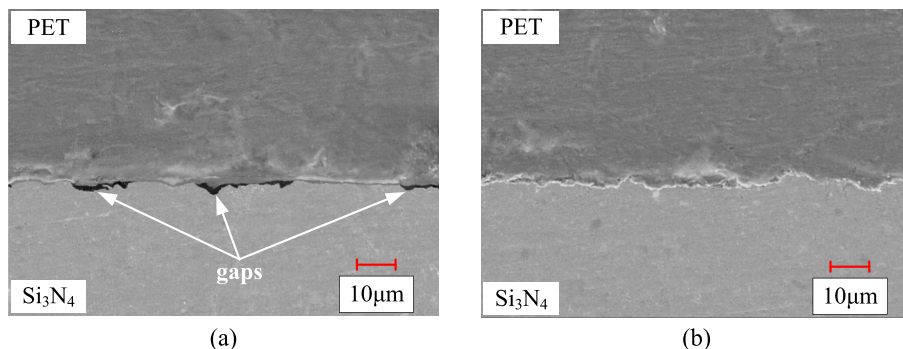


Fig. 5. Cross-sectional morphology of the interface between the PET and Si_3N_4 : (a) without ultrasonic vibration under $1000\times$, (b) with ultrasonic vibration under $1000\times$.

irrespective of the presence or absence of ultrasonic vibration. Specifically, the contact angle θ between the PET melt and Si_3N_4 is observed to be less than 90° , which indirectly corroborates the generation of interactive forces between PET molecules and Si_3N_4 atoms at the interface, thereby resulting in physical bonding. However, the wetting efficacy of PET on Si_3N_4 varies depending on whether ultrasonic vibrations are present or not. When ultrasonic vibrations are applied, the PET/ Si_3N_4 interface exhibits almost no discernible defects, and the PET melt thoroughly permeates the surface pits of Si_3N_4 as shown in Fig. 5(b). In contrast, in the absence of ultrasonic vibrations, noticeable gaps are apparent at the interface, with the pits being only partially filled by the PET melt as shown in Fig. 5(a). These observations indicate that ultrasonic vibration enhances the effective wetting of the Si_3N_4 surface by the PET melt during the laser joining process.

3.1.3. Enhancement of bonding area

Optical microscopy was employed to examine the joints produced by both laser joining (LACP) and ultrasound aided laser joining (U-LACP). Fig. 6 depicts the appearance of the joints and provides measurements of their bonding areas. For each welding condition, measurements were taken from three samples, with average values of 85.9 mm^2 and 73.4 mm^2 , respectively. The results indicate an approximate 17 % increase in the bonding area when ultrasonic vibration was applied during the laser joining process. According to our previous study [20], the increase in weld seam temperature caused by pulsed ultrasonic vibration can be considered negligible. In other words, the increase in the bonding area is not attributed to the generation of additional molten PET due to ultrasonic vibration. Moreover, minor changes in temperature that could affect the viscosity of the PET melt are also negligible. Chen [31] through rheological experiments, has demonstrated that the viscosity of polymer melts with non-Newtonian flow characteristics significantly decreases under the influence of ultrasonic vibrations. Both chemical and physical effects induced by ultrasonic vibration contribute to the reduction of viscosity in polymer melts. Ultrasonic irradiation can lead to random chain scission of polymer molecules, resulting in reduced

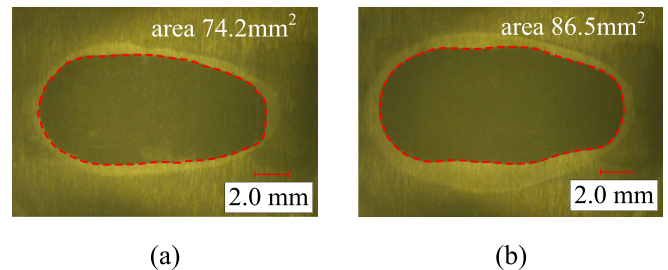


Fig. 6. Joint appearance examined by an optical microscope: (a) LACP specimen under $20\times$, (b) U-LACP specimen under $20\times$.

molecular weight and a narrower molecular weight distribution—these constitute the chemical effects that reduce the viscosity. Furthermore, physical effects manifest through increased molecular chain mobility and reduced elastic elongation strain, thereby facilitating viscosity reduction. Holtmannspotter et al. [22] and Wang et al. [32] found that reducing the viscosity of non-Newtonian adhesives through ultrasonic vibration significantly enhances their flowability, resulting in a rapid decrease in the adhesive's contact angle in a short period. High-speed cameras were used to document the spreading behavior of non-Newtonian adhesive droplets under ultrasonic vibrations. The research revealed that the droplets of non-Newtonian adhesive became thinner during spreading, a phenomenon attributed to forced wetting induced by ultrasonic vibration, which propels the adhesive to spread out. The molten PET also exhibits non-Newtonian flow behavior. Therefore, on one hand, the viscosity of the PET melt decreases under the influence of ultrasonic vibration in the laser joining process, enhancing the flowability of PET. Consequently, the contact angle formed between the molten PET and Si_3N_4 decreases. The molten PET more readily flows and penetrates into the rough Si_3N_4 surface before solidification, improving the wetting effect of molten PET on the Si_3N_4 surface. On the other hand, the forced wetting induced by ultrasonic vibration encourages the molten PET to spread out, thus enlarging the bonding area.

3.2. Improvement of chemical bonding at the PET/ Si_3N_4 hybrid joint interface

3.2.1. Thermodynamics conditions for chemical bonding at the laser joining interface between PET and metal or ceramic

The chemical bonding process between PET and metal or ceramic should conform to the basic principles of chemical reaction thermodynamics. Assuming that metal or ceramic can react with PET to form new chemical bonds during laser joining, the corresponding chemical reaction equation can be expressed as follows:



PET is primarily composed of carbon (C) and oxygen (O) elements, both of which may react with metals or ceramics (denoted as Me) to form Me-C or Me-O products. Therefore, the general equation for the chemical reaction between the metal and the C or O elements in PET at the bonding interface can be written as follows:

$$\Delta rG : \langle \text{Me} \rangle_{\text{ceramic}} + \langle \text{C or O} \rangle_{\text{PET}} = \langle \text{Me} - \text{C or Me} - \text{O} \rangle_{\text{int}} \quad (9)$$

In Eq. (9), ΔrG represents the variation in the Gibbs free energy of the chemical reaction at the interface between PET and Me, with indicating the solid state. The subscripts ceramic, PET and int refer to ceramic or metal sheets, PET sheets and their interface, respectively.

When calculating the variation in the Gibbs free energy (ΔrG) of the chemical reaction between PET and Me at their bonded interface, it is found that when $\Delta rG < 0$, the chemical reaction described by Eq. (9) can occur spontaneously. In other words, PET sheets can form chemical bonds with the metal or ceramic.

Eq. (9) represents a chemical reaction in the standard state, and the value of ΔrG cannot be directly calculated using the Gibbs-Helmholtz equation [27]. For further research, it is going to divide the chemical reaction between PET and Me at their bonded interface into two steps. The first step involves transforming the C or O in PET into their standard states,

$$\Delta rG_1 : \langle \text{C or O} \rangle_{\text{PET}} = (\text{C or } \frac{1}{2}(\text{O}_2)) \quad (10)$$

In Eq. (10), () indicating gas state.

The second step entails considering the reaction of metal or ceramic with the C or O in PET when they are in their standard states,

$$\Delta rG_2 : \langle \text{Me} \rangle_{\text{ceramic}} + (\text{C or } \frac{1}{2}(\text{O}_2)) = \langle \text{Me} - \text{C or Me} - \text{O} \rangle_{\text{int}} \quad (11)$$

The Gibbs function variation (ΔrG) of chemical reaction of PET and metal or ceramic at their laser bonded interface equals to the sum of ΔrG_1 , the Gibbs function variation of the chemical reaction in which C or O in PET transitions to a standard state, and ΔrG_2 , the Gibbs function variation of chemical reaction between Me and C or O in standard state. Therefore, the relationship among ΔrG , ΔrG_1 and ΔrG_2 is,

$$\Delta rG = \Delta rG_1 + \Delta rG_2 \quad (12)$$

Therefore, it has to calculate ΔrG_1 and ΔrG_2 to know ΔrG . As the target, the polymer material PET and laser joining parameters (laser power, joining speed, pulse frequency and duration etc.) are confirmed, $\Delta rG_1(\text{PET})$ is a fixed value, while in ΔrG_2 , as the metal or ceramic is not defined and $\Delta rG_2(\text{Me})$ changes as the metal or ceramic varies. Now take Me as a specific one, then,

$$\Delta rG_2(\text{Me}) < -\Delta rG_1(\text{PET}) \quad (13)$$

When the Gibbs function varies in the chemical reaction between PET and a metal or ceramic at their bonded interface $\Delta rG < 0$, the metal or ceramic (Me) can form a chemical bond with PET. Therefore, Eq. (13) represents the thermodynamic condition for the formation of a chemical bond between a metal or ceramic (Me) and PET, where the key thermodynamic parameter to be determined is $\Delta rG_1(\text{PET})$.

However, the value of $\Delta rG_1(\text{PET})$ cannot be directly determined through either experimental or theoretical calculations. Nonetheless, studies [33–40] have shown that commonly used commercial metals or alloys, such as titanium, aluminum, magnesium, copper, and iron, can undergo chemical reactions with PET during the laser joining process, forming new Me-C or Me-O chemical bonds. Based on the above analysis, we can conclude that,

$$\begin{aligned} \Delta rG(\text{Cu or Ti or Fe or Al etc}) &= \Delta rG_1(\text{PET}) + \Delta rG_2(\text{Cu or Ti or Fe or Al etc}) \\ &< 0 \end{aligned} \quad (14)$$

With deduction, it can be known that $\Delta rG_1(\text{PET})$ satisfies:

$$\Delta rG_1(\text{PET}) \leq -\Delta rG_2(\text{Cu or Ti or Fe or Al etc.}) \quad (15)$$

During laser joining, the premise of chemical bonding between PET and metal or ceramic is that PET melts and wets the metal or ceramic surface, so the lowest temperature of interfacial chemical reaction is about 260 °C (533 K) of PET melting temperature. And, in this work, the temperature at the center of the weld during joining is around 305 °C (578 K) captured by a thermal infrared imager, which is sufficient to melt the PET material. The Gibbs-Helmholtz equation [27] was used to solve the thermodynamic parameters of the second-step chemical reaction (Eq. (11)) of Me, which is expressed as,

$$\Delta rG_2(\text{Me}) = \Delta H_2 - T\Delta S_2 \quad (16)$$

ΔrH_2 is the standard molar enthalpy change of the second-step chemical reaction, and ΔrS_2 is the standard molar entropy change of the second-step chemical reaction. The thermodynamic parameters related to the chemical bonding of common metal elements with PET at the laser-joining interface is presented in Table 2 [41]. The calculated $\Delta rG_2(\text{Cu or Ti or Fe or Al etc.}, 578 \text{ K})$ then,

$$\Delta rG_1(\text{PET}) \leq \Delta rG_2(\text{Cu}) = 24.3 \text{ kcal/mol} \quad (17)$$

Substituting Eq. (17) into Eq. (13), it can be seen that when $\Delta rG_2(\text{Me}, 578 \text{ K}) < -24.3 \text{ kcal/mol}$, $\Delta rG < 0$, then the metal Me can react with PET to form new chemical bonds during laser joining. In other words, the thermodynamic condition for a chemical reaction to occur at the interface during the laser joining PET to metal or ceramic (Me) is $\Delta rG_2(\text{Me}, 578 \text{ K}) \leq -24.3 \text{ kcal/mol}$.

3.2.2. XPS analysis results of the interface between PET and Si_3N_4 joint

The XPS analysis was performed on the joint interface that was ion-milled to varying depths from the Si_3N_4 surface after mechanical

Table 2

The thermodynamic parameters related to the interface chemical reaction between common metal and PET during laser joining [41].

Thermodynamic parameter	Fe	Cu	Ni	Zn	Al	Mg	Cr	Ti
$\Delta_f H_2$ (298 K)	−88.96	−37.1	−57.50	−83.80	−200.45	−143.70	−135.00	−129.70
$\Delta_f S_2$ (298 K)	−27.12	−22.22	−22.54	−24.02	−37.42	−25.87	−32.70	−23.52
$\Delta_f G_2$ (533 K)	−74.51	−25.26	−45.49	−71.00	−180.51	−129.91	−117.57	−117.16
$\Delta_f G_2$ (578 K)	−73.34	−24.30	−44.52	−69.96	−178.90	−128.80	−116.16	−116.15

separation. The samples were ion-milled to a depth just before the elements of Si or Ti been detected (designated $d = 0$ or $d_u = 0$). Figs. 7 and 8 show the high-resolution N1s and Si2p XPS spectra obtained from samples at different milling depths, produced by direct laser joining (LACP) and ultrasonic-assisted laser joining (U-LACP), respectively. The results indicate that the chemical composition and state in this region are similar to those of the original Si_3N_4 sheet, and no peaks related to carbon bonds were detected. This suggests that no chemical reactions occurred between Si_3N_4 and PET during the laser joining process, even when assisted by ultrasonic vibration. The results also show that the variation in the Gibbs function for the chemical reaction between Si_3N_4 and C or O elements in standard state is $\Delta_r G_2(\text{Si}_3\text{N}_4, 578 \text{ K}) > -24.3 \text{ kcal/mol}$, even with the aided of ultrasonic vibrations.

3.2.3. XPS analysis results of the interface between PET and Ti-coated Si_3N_4 joint

In prior study [39], a new chemical bond Ti—C was detected at the joint interface, characterized by a binding energy of approximately 281.7 eV. To further investigate this, ion milling was conducted on the samples at various depths, aiming to confirm the existence of Ti—C bond at PET/Ti-coated Si_3N_4 interface. High-resolution XPS spectra of the C1s lines were obtained for both LACP and U-LACP specimens, with different milling depths. For the LACP specimen, the peaks corresponding to the Ti—C bond disappeared at a milling depth of approximately 700 nm, indicating a thickness of around 700 nm for the interface containing the Ti—C bond. However, for the U-LACP specimen, the Ti—C peak was still visible at a depth of about 1100 nm, as shown in Fig. 9. This suggested an interface thickness that was approximately 1.57 times of that found in the LACP samples. This could be largely attributed to the ultrasonic vibration enhances the collision and adsorption between carboxyl group carbon (C) atoms and Ti atoms at the bonding interface. This, in turn, led to the liberation of additional thermal energy, intensifying the heat effects and resulting in a decrease in the enthalpy change (ΔH) of the reaction [42]. This increased thermal activity improved the propensity for chemical reactions to occur. Furthermore, the active metal titanium (Ti) demonstrates strong electrophilicity due to its capacity to lose electrons and bear a positive charge. Consequently, electrophilic Ti + species at

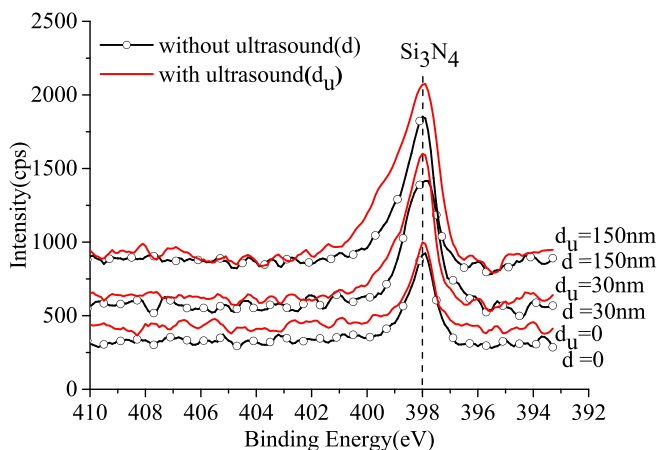


Fig. 7. High resolution N1s XPS spectrum taken from Si_3N_4 side surface for both LACP and U-LACP samples with different milling depths.

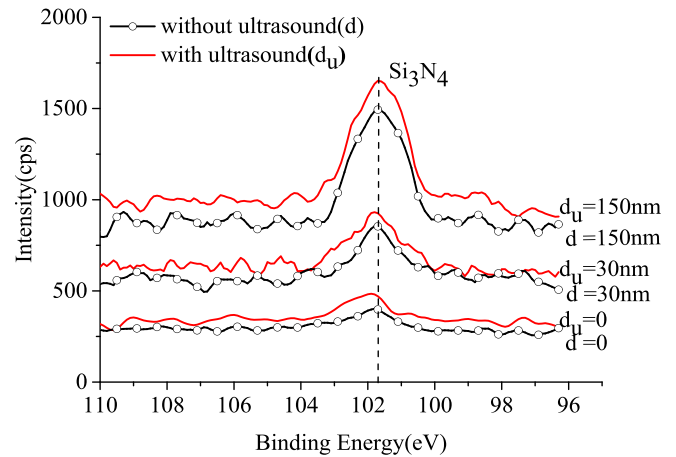


Fig. 8. High resolution Si2p XPS spectrum taken from Si_3N_4 side surface for both LACP and U-LACP samples with different milling depths.

the interface exhibit an elevated probability of attacking nucleophilic C=O groups on the carboxyl, promoting reactions to form Ti—C bonds. Finally, changes in internal pressure, triggered by ultrasonic vibrations, could affect the entropy changes within the system [42]. As the ultrasonic waves coursed through the molten PET, they induced high-frequency molecular oscillations. These vibrations led to increased molecular activity and collision frequency, facilitating easier molecular rearrangement and thereby increasing the system's entropy change (ΔS). According to the Gibbs-Helmholtz equation, these synergistic effects led to a decrease in the variation of the Gibbs function ($\Delta_r G$), collectively promoting chemical reactions between titanium atoms and carbonyl groups, ultimately resulting in the formation of new Ti—C bonds.

3.3. Improvement of mechanical interlocking at the PET/ Si_3N_4 hybrid joint interface

The micro-blind hole structures on the Si_3N_4 ceramic plate, the starting point, end point, joining direction and trajectory of joining are indicated in Fig. 10(a). The joint is sectioned along the xOz plane as shown in the same figure. The profiles of cross section were observed by optical microscope and the filling depth of molten PET were measured, the results are displayed in Fig. 10(b) and (c). It can be observed that the joint obtained through laser joining has melted PET filling in the blind holes ranging from H4 to H11 along the joining direction. In the case of ultrasonic assisted laser joining, the blind holes of H3-H12 are filled with molten PET. This suggests that ultrasonic vibration enlarges the bonding area. Additionally, it can be observed that the depth of melted PET fill in samples produced through laser joining is around 520 μm . However, when ultrasonic vibrations were applied during the laser joining process, the depth of melted PET filling increased to approximately 900 μm , representing an increase of roughly 73 %. During the laser joining process, the molten PET flows into the blind holes under the combined influence of gravity, clamping force, and capillary action. Furthermore, Wang et al. [32] conducted the ultrasonic capillary experiment, and found that with the progress of ultrasonic vibration, the polymer adhesive will also be affected by a driving force, which drives the polymer

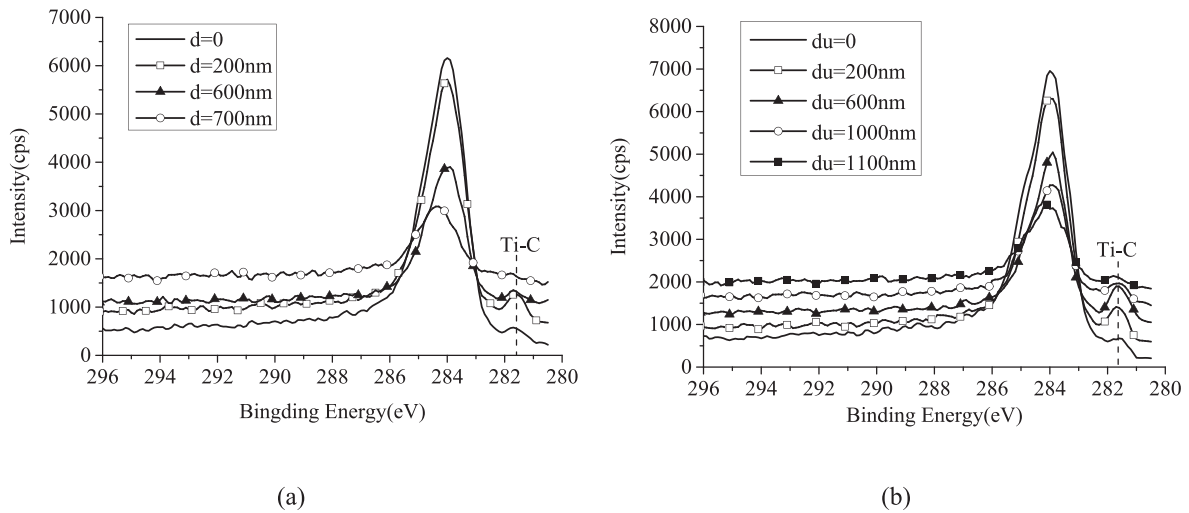


Fig. 9. High resolution Ti2p XPS spectrum taken from Ti-coated Si_3N_4 surface with different milling depths: (a) LACP specimen, (b) U-LACP specimen.

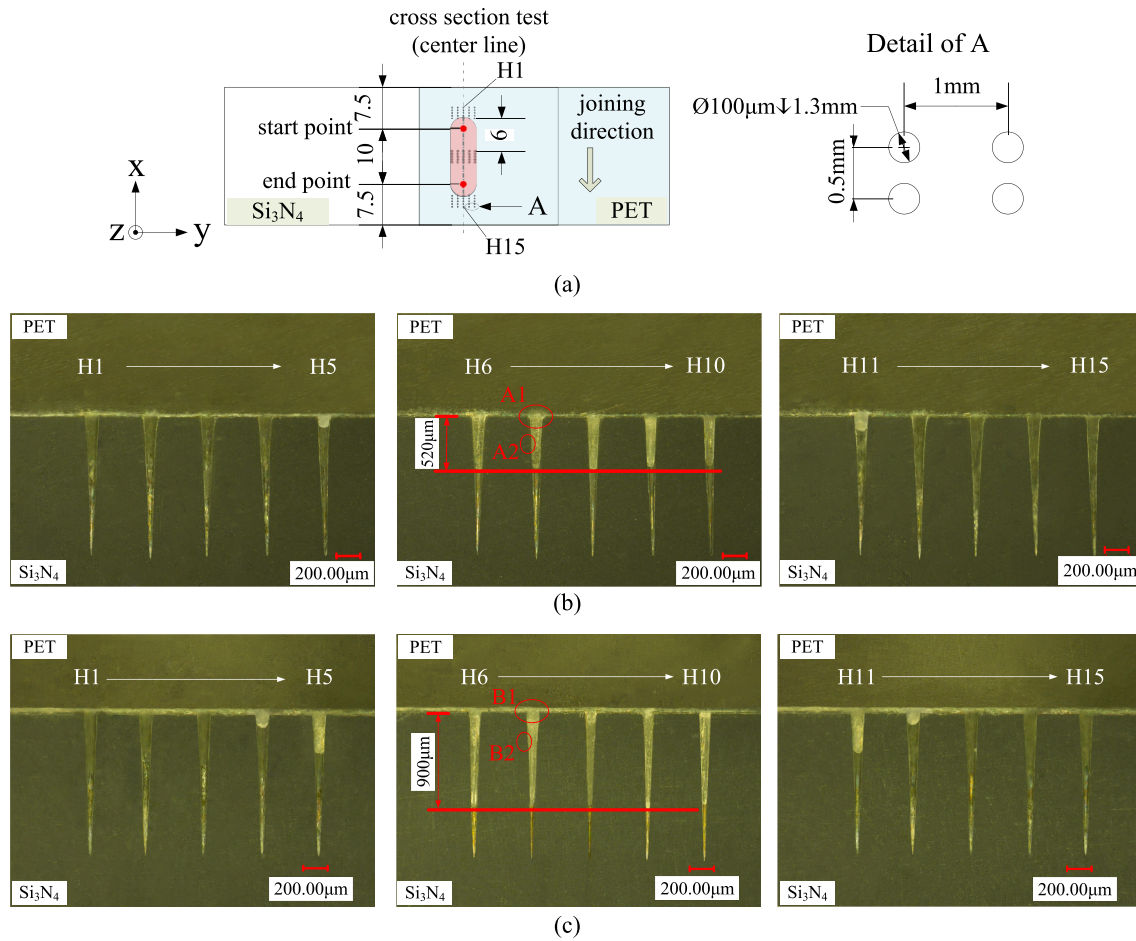


Fig. 10. (a) a schematic illustration showing the micro-blind hole structures on the Si_3N_4 ceramic plate and the joining process; cross section diagram of the joints along the joining direction: (b) without ultrasonic vibration, (c) with ultrasonic vibration.

adhesive to fill in the tube. Hence, in ultrasonic-assisted laser joining, a greater volume of molten PET flows into the blind holes.

To further study the promoting effect on mechanical interlocking, the blind holes H7 in the joints produced by laser joining and ultrasound aided laser joining were observed and analyzed by the SEM. The bonding morphology of the PET and Si_3N_4 at the entrance (position A1

and B1 as shown in Fig. 10) and the wall (position A2 and B2 as shown in Fig. 10) of the blind holes H7 are shown in Fig. 11(a)–(d). In the absence of ultrasonic vibration, obvious gaps can be seen at the entrance and wall of the blind hole as shown in Fig. 11(a) and (c), respectively. When ultrasonic vibration is applied, PET forms a tight bond both at the entrance and the wall of the blind hole, as shown in Fig. 11(b) and (d).

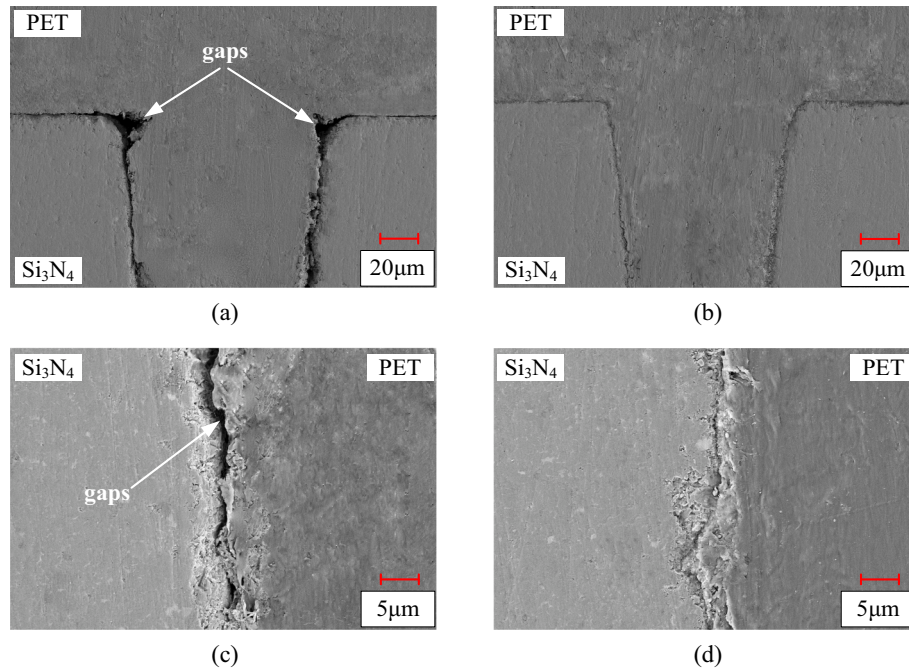


Fig. 11. The bonding morphology of the entrance and the wall of the blind holes H7 in the joints produced by laser joining and ultrasound aided laser joining: (a) position A1 under 500×, (b) position B1 under 500×, (c) position A2 under 2000×, (d) position B2 under 2000×.

On one hand, as mentioned above, ultrasonic vibration leads to an obvious reduction in the viscosity of the PET melt results in a significant improvement of fluidity, which makes molten PET flow and permeate into rough entrance and blind hole walls more easily. On the other hand, the longitudinal vibrational ultrasonic is applied in this research, which direction is essentially parallel to the direction of melt flow and the direction of blind hole depth. It will cause the reciprocating movement of the PET melt in the blind hole, resulting in friction between the PET melt and both the wall and entrance of the blind holes. As a result, a tighter bond was formed between the PET melt and the Si₃N₄ ceramic.

Therefore, the mechanism of ultrasonic vibration improving mechanical interlocking can be illustrated by Fig. 12. Firstly, in addition to the effects of gravity, extrusion force, and capillary force, the PET melt is also subjected to a driving force, which drives the molten PET filled into the blind holes before it was cured. Secondly, the viscosity of the molten PET was decreased under the ultrasonic vibration, the fluidity of the PET melt increased. As a result, the molten PET flow and permeate into

rough entrance and blind hole walls more easily. Finally, owing to the ultrasonic-induced friction between the PET melt and the blind hole wall and entrance, gap defects were reduced significantly. Therefore, the mechanical interlocking was obviously promoted.

4. Conclusion

This study successfully achieved the bonding of Si₃N₄ plates to PET and Ti-coated Si₃N₄ plates to PET by laser transmission joining. Analytical techniques such as SEM, XPS, chemical bond thermodynamics, and wetting thermodynamics were employed to elucidate the bonding mechanism at the interface and the enhancement principle of ultrasonic vibration on interface bonding. The primary findings are summarized as follows:

1. Whether using ultrasound assistance or not, the interface between laser-joined Si₃N₄ and PET exhibits only physical and mechanical

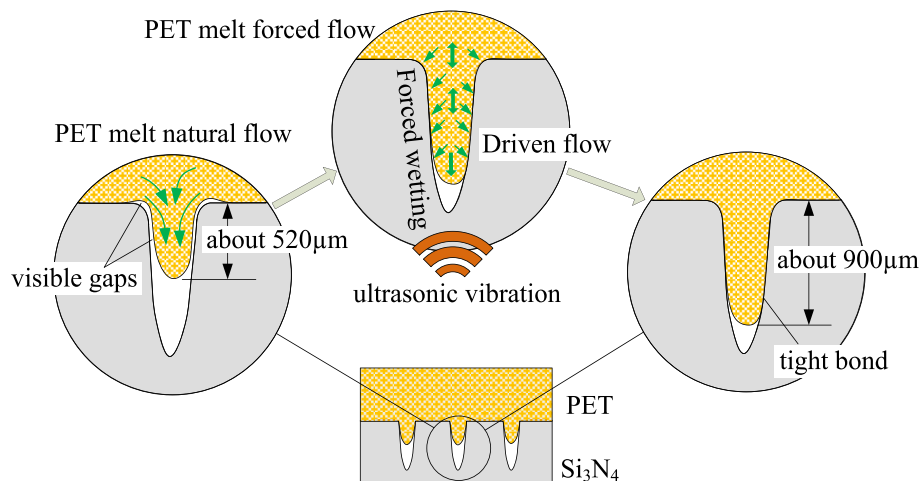


Fig. 12. Schematic diagram of ultrasonic promoting mechanical interlocking at the blind holes.

- bonding, with no evidence of new chemical bonds. However, XPS analysis detected new Ti—C bonds at the interface of Ti-coated Si₃N₄ connections.
- The physical wetting thermodynamic conditions and chemical bond thermodynamic conditions for laser joining of PET to ceramics or metals were the surface tension of ceramics or metals is higher than that of the molten PET and $\Delta rG_2(\text{Me}, 578\text{ K}) \leq -24.3\text{ kcal/mol}$, respectively.
 - In terms of mechanical bonding, on one hand, ultrasonic vibration increases the friction between PET melt and rough blind hole, on the other hand, it reduces the viscosity of PET melt and increases its fluidity, making it easier for PET melt to flow into the blind hole and pass through the rough blind hole entrance and hole wall to form mechanical anchoring. The ultrasonic vibration generated additional driven force in the PET melt, which increased the filling depth of the blind hole by about 73 %.
 - Concerning chemical bonding, ultrasonic vibrations intensified the collision and friction between atoms, promoting the chemical reaction between the molten PET and Ti atoms. This interaction thickened the interface containing the Ti—C bonds by roughly 57 %.
 - From the perspective of physical bonding, ultrasonic vibrations ensured the complete penetration of molten PET into the recesses of the Si₃N₄ surface, leading to a tighter contact between the PET melt and Si₃N₄. Furthermore, the forced wetting induced by ultrasonic vibrations pushed the molten PET to spread outwards, expanding the bonding area by approximately 17 %. Compared to mechanical and chemical bonding, the enhancement in physical bonding was less pronounced.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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